Euclidean Geometry In Mathematical Olympiads

Problem 1.46 (Canada 1997/4)

728 posts

Incomplete sol (stuck)

franzliszt
23527 posts

Jul 18, 2022, 2:41 PM #13

Let O' be the point such that $ABOO'$ is a parallelogram. Since $AB = OO' = CD$ and $AB \parallel OO' \parallel DC$, we find that $DCOO'$ is also a parallelogram. Using directed angles, the condition rewrites to $\measuredangle BOA = \measuredangle COD$. However, observe $\measuredangle BOA = \measuredangle O'AO$ and $\measuredangle COD = \measuredangle O'DO$ so $AODO'$ is cyclic. Now notice that $\measuredangle CDO = \measuredangle O'OD = \measuredangle O'AD = \measuredangle OBC$ so we are done.

cinnamon_e
704 posts

May 25, 2023, 7:06 PM #14

solution

Translate the triangle COD so that AB and DC are the same line segment, and O is outside the parallelogram $ABCD$. Call the translated triangle $C'O'D'$.
Note that because $\angle COD + \angle AOB = 180^\circ$, then $\angle AO'B + \angle AOB = 180^\circ$, so $AOBO'$ is cyclic.
Hence,

$$\angle ODC = \angle O'AB = \angle O'OB,$$

and because $OO' \parallel BC$, $\angle O'OB = \angle OBC$, so we are done.

pqr.
174 posts

Jun 6, 2023, 6:43 PM #15

Pretty cool problem, but *extremely* difficult.

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Construct O' such that $AOO'D$ and $BOO'C$ are parallelograms. Then, notice that $CODO'$ is cyclic because $\angle CO'D + \angle COD = 180^\circ$. Then, using properties of cyclic quadrilaterals, we see that

$$\angle OBC = \angle CO'O = \angle ODC,$$

as desired.

This post has been edited 2 times. Last edited by pqr., Jun 6, 2023, 6:44 PM

BakedPotato66
747 posts

Jan 1, 2024, 6:35 PM #16

fjm30 wrote: